Study Studio — AI Cost Sheet

Benchmark of the production build pipeline · 8 October 2026 · prices from OpenRouter on the same day

1. Summary

We built two real studios in the production app — a single 9-page PDF reading, and a mixed set of two textbook chapters, a photographed page and a slide deck — once with each build model. Every number below is the cost OpenRouter reported for each call, with model "thinking" turned off.

	Standard build Gemini 3.1 Flash Lite	**Deep build GLM 5.3 Flash**
Single PDF (9 pages, 31k chars)	$0.0276 · 7 lessons · 17 s	$0.0349 · 19 lessons · 6.2 min
2 PDFs + image + PPTX (63 pages, 186k chars)	$0.0438 · 10 lessons · 22 s	$0.0385 · 17 lessons · 6.9 min
Cost per lesson	$0.0029 – $0.0027	$0.0016 – $0.0017
Lesson depth (avg. explanation length)	1,020 – 1,012 chars	2,096 – 3,149 chars

Key points. A studio build costs about **3–5 cents** with either model. Ask AI costs about **$0.009** an answer, and grading a spoken recall answer costs a hundredth of a cent. Uploading and reading files (text extraction, OCR, search index) uses no paid AI. GLM 5.3 Flash writes **2–3× more lessons**, each **2–3× longer**, for a similar total cost, but a build takes **6–7 minutes** instead of about 20 seconds — so we offer it as a **Deep build** (better studio, longer wait) next to the fast **Standard build**.

2. How it was measured

• **Real pipeline:** the production Rust service (study-api) on the production server, with its own upload, ingest and build workers, run against throwaway copies of the production databases (no user data changed).
• **One run per configuration** (2 materials × 2 build models). AI output length varies between runs; expect roughly ±20–30%.
• **Thinking off** on every call. GLM 5.3 Flash cannot turn thinking off (OpenRouter returns "Reasoning is mandatory"), so it runs with *effort: minimal*: 16–159 thinking tokens per whole build.
• **Costs** are OpenRouter's own *usage.cost* per call, stored per call in *llm_usage* (tokens in/out, thinking tokens, cost, latency, build id).
• After each build: 3 Ask AI questions and 3 spoken-recall gradings on the new studio.

3. Test material

Run	File	Type	Size	Pages	Text read	Chunks	Ingest
Build 1	EBSCO-FullText-09_08_2026.pdf	PDF (text)	2.57 MB	9	30,886 chars	32	38 s
Build 2	marketing-ch1.pdf	PDF (text)	1.44 MB	27	76,873 chars	90	47 s
Build 2	marketing-ch2.pdf	PDF (text)	1.82 MB	36	107,456 chars	118	58 s
Build 2	marketing-ch3-page.jpg	Image (OCR)	0.32 MB	1 photo	3,138 chars	2	9 s
Build 2	marketing-ch3-slides.pptx	PowerPoint	0.03 MB	5 slides	1,966 chars	2	2 s

Build 1 is a 9-page journal reading. Build 2 is OpenStax *Principles of Marketing* (CC BY): chapter 1 and chapter 2 as PDFs, a photo of the first page of chapter 3 (JPG, read by OCR), and a 5-slide deck on chapter 3. Ingest — text extraction, OCR (self-hosted Hirara), splitting into passages and embedding (self-hosted bge-m3) — costs no AI money. In the Flash Lite run of Build 2 the photo's OCR came back empty while the server was busy (fixed since: OCR now retries); it read correctly in the GLM run.

4. Models and prices (OpenRouter, USD per 1M tokens)

Model	Used for	Input	Output	Cached input	Max output	Thinking
Gemini 3.1 Flash Lite google/gemini-3.1-flash-lite-preview	Standard build, grading	$0.250	$1.50	$0.025	65,536	off
GLM 5.3 Flash z-ai/glm-5.3-flash	Deep build (tested)	$0.150	$0.50	$0.030	943,717	can't turn off
Claude Sonnet 4.6 anthropic/claude-sonnet-4-6	Ask AI, concept maps	$3.000	$15.00	$0.300	64,000	off

5. Build results

	B1 · Flash Lite	B1 · GLM	B2 · Flash Lite	B2 · GLM
Material read	30,886 chars	30,886 chars	186,295 chars	189,433 chars
Lessons	7	19	10	17
Flashcards / quiz questions	49 / 28	173 / 114	72 / 40	168 / 102
Scenarios	5	7	0	9
Lesson text written	7,146 chars	39,832 chars	10,123 chars	53,533 chars
AI calls	9	21	12	19
Tokens in / out	32,931 / 12,901	68,188 / 49,963	71,631 / 17,282	85,598 / 51,849
Thinking tokens	0	16	0	159
Build time	17 s	6.2 min	22 s	6.9 min
Build cost	**$0.0276**	**$0.0349**	**$0.0438**	**$0.0385**

Build 2 with Flash Lite returned no scenarios: the model's reply couldn't be parsed (now logged for diagnosis). Its cost still includes that call.

Cost by call type

Call	What it reads	Runs per build	B1 · FL	B1 · GLM	B2 · FL	B2 · GLM
Plan the lessons	All material, up to 120,000 chars	1	$0.0025 1× · 7,225 in / 480 out	$0.0017 1× · 7,053 in / 1,290 out	$0.0072 1× · 24,918 in / 649 out	$0.0042 1× · 24,269 in / 1,158 out
Write one lesson	8 most relevant passages	1 per lesson	$0.0204 7× · 18,389 in / 10,501 out	$0.0304 19× · 53,989 in / 45,133 out	$0.0274 10× · 21,703 in / 14,618 out	$0.0284 17× · 36,967 in / 46,220 out
Application scenarios	All material, up to 120,000 chars	1	$0.0047 1× · 7,317 in / 1,920 out	$0.0028 1× · 7,146 in / 3,540 out	$0.0093 1× · 25,010 in / 2,015 out	$0.0059 1× · 24,362 in / 4,471 out

Lessons (each also gets its flashcards, quiz and concept map in the same call) are **62–88%** of a build's cost. The outline and scenario calls read the material itself, so they grow with its size until the 120,000-character cap.

6. Other features

Feature	Model	Tokens per call (in / out)	Cost per call	Notes
Ask AI (one answer)	Claude Sonnet 4.6	1,945 / 222	$0.0092	12 answers measured, $0.0087–0.0097

Feature	Model	Tokens per call (in / out)	Cost per call	Notes
Spoken recall grading	Build model	145 / 27	$0.00004 – 0.00007	Only when the word check fails
Concept map (older studios)	Claude Sonnet 4.6	—	≈ $0.017*	New builds include maps at no extra call
Topic connections (older studios)	Claude Sonnet 4.6	—	small*	New builds include them
Upload / ingest / search	self-hosted	—	$0 AI	Server CPU only
Copy a shared studio	—	—	$0 AI	Vectors are copied

* measured on 7 Oct 2026 (connections + one concept map = $0.017 together).

7. Cost model — how to estimate any build

build cost ≈ outline(material) + scenarios(material) + lessons × cost per lesson

• **outline + scenarios** read the whole material (≈ 4 characters per token), capped at 120,000 characters (≈ 25,000 tokens) each.
• **lessons** — the model decides how many from the material: a short note 2–3, a chapter 6–12, a long book 20+. GLM splits finer (2–3× as many as Flash Lite on the same material).
• **cost per lesson** is steady across materials (each lesson reads ~8 passages).

Constant	Standard (Flash Lite)	Deep (GLM 5.3 Flash)
Outline + scenarios, small material (31k chars)	$0.0072	$0.0045
Outline + scenarios, at the 120k cap	$0.0165 measured · ≤ $0.0165	$0.0101 measured · ≤ $0.0103
Cost per lesson	$0.0029 – $0.0027	$0.0016 – $0.0017
Tokens per lesson (in / out)	2,627 / 1,500	2,842 / 2,375

Estimated builds by size

Material	Standard lessons	Standard cost	Deep lessons	Deep cost
Short note (1–3 pages, ~6k chars)	3	$0.013	6	$0.013
Reading (9 pages, ~31k chars)	7	$0.027	19	$0.036
Chapter (~40 pages, ~110k chars)	11	$0.048	24	$0.051
Several chapters (≥120k chars, capped)	15	$0.060	30	$0.062
Whole book (cap reached, many topics)	25	$0.087	40	$0.079

Row 2 matches Build 1 and row 4 is close to Build 2 (measured); the lesson counts in the other rows are estimates. Today the outline reads only the first 120,000 characters, so very large uploads get lessons for their first ~40–50 pages only — raising the cap would add about $0.01–0.02 per large build plus the extra lessons.

8. One build with other models (projected)

Measured token counts × each model's price. The Standard rows use Flash Lite's tokens (fewer, shorter lessons); the Deep rows use GLM's tokens (more, longer lessons). A different model will write a different amount, and quality and speed are untested — these are price comparisons, not recommendations.

Model	In / out per 1M	Standard-size build B1 · B2	Deep-size build B1 · B2	Ask AI answer
Gemini 3.1 Flash Lite (Standard measured) google/gemini-3.1-flash-lite-preview	$0.250 / $1.50	$0.0276 · $0.0438	$0.0920 · $0.0992	$0.00082

Model	In / out per 1M	Standard-size build B1 · B2	Deep-size build B1 · B2	Ask AI answer
GLM 5.3 Flash (Deep measured) z-ai/glm-5.3-flash	$0.150 / $0.50	$0.0114 · $0.0194	$0.0352 · $0.0388	$0.00040
Claude Haiku 5.5 anthropic/claude-haiku-5.5	$0.100 / $0.50	$0.0097 · $0.0158	$0.0318 · $0.0345	$0.00031
GPT-6 Luna openai/gpt-6-luna	$0.100 / $0.50	$0.0097 · $0.0158	$0.0318 · $0.0345	$0.00031
Qwen3.8 Flash qwen/qwen3.8-flash	$0.150 / $0.47	$0.0110 · $0.0189	$0.0337 · $0.0372	$0.00040
MiMo-V2.6-Flash xiaomi/mimo-v2.6-flash	$0.140 / $0.28	$0.0082 · $0.0149	$0.0235 · $0.0265	$0.00033
Gemini 2.5 Flash Lite google/gemini-2.5-flash-lite	$0.100 / $0.40	$0.0085 · $0.0141	$0.0268 · $0.0293	$0.00028
DeepSeek V4 Pro 0423 deepseek/deepseek-v4-pro	$0.292 / $0.58	$0.0172 · $0.0310	$0.0491 · $0.0553	$0.00070
gpt-oss-120b openai/gpt-oss-120b	$0.037 / $0.17	$0.0034 · $0.0056	$0.0110 · $0.0120	$0.00011
Qwen3.7 Flash qwen/qwen3.7-flash	$0.030 / $0.13	$0.0027 · $0.0044	$0.0085 · $0.0093	$0.00009
Gemini 3.8 Flash google/gemini-3.8-flash	$0.750 / $3.75	$0.0731 · $0.1185	$0.2385 · $0.2586	$0.00229
Claude Sonnet 4.6 (Ask AI today) anthropic/claude-sonnet-4-6	$3.000 / $15.00	$0.292 · $0.474	$0.954 · $1.035	$0.00916 (measured)

Measured: Flash Lite's Standard columns and GLM's Deep columns (OpenRouter's billed cost) and Sonnet's Ask AI. Every other cell is a projection that ignores cache discounts and retries.

9. Credit economics

OctoCredit packs sold through Stripe (fee assumed 2.9% + $0.30 per payment).

Pack	Price	Price per credit	Net per credit after Stripe
50 credits	$0.99	$0.0198	$0.0132
300 credits	$4.99	$0.0166	$0.0152
500 credits	$7.99	$0.0160	$0.0149
1,000 credits	$9.99	$0.0100	$0.0094
2,500 credits	$19.99	$0.0080	$0.0076
5,000 credits	$39.99	$0.0080	$0.0077

The cheapest credit we sell nets **$0.0076** (the 5,000 pack). Margins below use that worst case; smaller packs earn more per credit.

Suggested prices

Action	Credits	AI cost typical / high	Revenue at worst-case credit	Margin typical / high
Standard build (Flash Lite, ~20 s)	25	$0.0357 / $0.0874	$0.191	5.4× / 2.2×
Deep build (GLM 5.3 Flash, 6–7 min)	40	$0.0367 / $0.0786	$0.306	8.3× / 3.9×
Ask AI answer	5	$0.0092 / $0.0125	$0.038	4.2× / 3.1×
Concept map (older studios)	10	$0.0170 / $0.0250	$0.076	4.5× / 3.1×
Spoken recall grading	0 (free)	$0.0001 / $0.0001	—	free

"High" = a large upload at the cap with many lessons (25 Standard / 40 Deep). With the 1,000-credit pack (net $0.0094 a credit) every margin is about 25% higher than shown.

How to set them

Prices live in the octopilot database's *system_settings* and apply within a minute, no deploy:
• fixed per action: *study_price_build*, *study_price_ask*, *study_price_concepts*, *study_price_connections*, *study_price_grade* (0 = free);
• or metered: *study_credits_per_usd* — real OpenRouter cost × rate, rounded up, at least 1 credit (e.g. 500 ⇒ Standard build ≈ 14–22, Ask AI ≈ 5).
A user short of credits gets "not enough OctoCredits" before any AI call; builds are charged only when they succeed. A Deep build option needs a small app change (a build-mode choice → the GLM model) before it can be sold.

10. Caveats

• One run per configuration; lesson counts and lengths vary between runs (±20–30% expected).
• Two materials only (a reading and a textbook set). Very large uploads hit the 120k-character outline cap.
• Model prices are a snapshot of 8 Oct 2026; Claude Haiku 5.5 and GPT-6 Luna were days old and may change.
• GLM's thinking can't be switched off; at minimal effort it used 16–159 thinking tokens a build (negligible).
• Ask AI uses Claude Sonnet 4.6 — by far the most expensive per call; moving it to a cheaper model is the largest saving available (see section 8), quality to be checked first.
• Re-check monthly from *llm_usage* (every call: model, tokens, thinking tokens, cost, latency, feature, build).

Appendix — lessons produced

Build 1 · Flash Lite (7)

Defining Corporate Leadership · The Transparency Imperative · Candor and Performance · Barriers to Candor · The Inevitability of Transparency · Building a Culture · Institutionalizing Transparency

Build 1 · GLM 5.3 Flash (19)

New metric of leadership · Defining transparency · Candor and upward communication · Groupthink and denial · Board oversight failures · Why leaders resist transparency · Transparency is inevitable · Guidant concealment case · DaVita no-secrets policy · Tell the truth · Speaking truth to power · Rewarding contrarians · Practicing unpleasant conversations · Diversifying information sources · Admitting mistakes · Organizational support for candor · Boards as last line of defense · Setting information free · Leaders as role models

Build 2 · Flash Lite (10)

Marketing and Process · Marketing Mix Fundamentals · Marketing Environment Factors · Evolution of Marketing · Consumer Needs and Wants · Customer Relationship Management · Ethical Marketing · Strategic Planning Process · Marketing Strategic Role · Marketing Strategy Analytics

Build 2 · GLM 5.3 Flash (17)

Defining Marketing · Marketing benefits and utility · The Marketing Process · Customer value and value proposition · The Marketing Mix and 4Ps · Internal Marketing Environment · Microenvironment Factors · Macroenvironment and PESTLE · Evolution of Marketing Concepts · Determining Consumer Needs and Wants · Customer Relationship Management · Ethical Marketing · Strategic Planning Fundamentals · Gap Analysis and SMART Goals · Marketing in Strategic Planning · SWOT Analysis · Consumer Buying Behavior